Pranav Gujjar

AI/ML Engineer |Computer Vision | Generative AI | Production AI Systems | MLOps

Bengaluru, India mr.pranavgujjar@gmail.com +91-9008913366 LinkedIn GitHub Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with an MSc in Data Science (Distinction, University of East Anglia, UK) and hands-on experience developing production-ready AI systems across Computer Vision, Machine Learning, NLP, and Generative AI. Proven expertise in designing end-to-end AI applications—from data pipelines and model development to deployment using FastAPI, MediaPipe, Docker, Streamlit, GitHub Actions, and cloud platforms. Delivered AI solutions that reduced manual analysis by 80%+, accelerated business decision-making by 50%, and improved operational efficiency through scalable, production-grade machine learning systems.

CORE TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, scikit-learn, PyTest), SQL, PySpark, R, MATLAB
ML / DL: Classification, Regression, Clustering, CNNs, RNN, LSTM, GRU, Transformers, TensorFlow, PyTorch
MLOps & Deployment: FastAPI, REST APIs, Docker, GitHub Actions CI/CD, Model Versioning,Artifact Management, Reproducible ML Pipelines, Streamlit.
Computer Vision: MediaPipe, Pose Estimation,Biomechanics Analysis,Motion tracking,OpenCV, U-Net, Dense U-Net, Grad-CAM, Image Segmentation, CLAHE, Patch-Based Learning, IoU, Dice
NLP & Speech: TF-IDF, Linear SVM, Whisper ASR, Rule-Based NLP, Sentiment Analysis, Intent Classification
Generative AI: LLMs,Prompt Engineering,AI Agents,GenAI Workflows,AI-assisted Coaching,RAG
Data & Analytics: PostgreSQL, MongoDB, Tableau, Power BI, Matplotlib, A/B Testing, Feature Engineering
Cloud / Infra: Render, Vercel, Streamlit Cloud (deployment), GitHub Actions

EDUCATION

MSc in Data Science — Distinction · University of East Anglia, UK *Sept 2023 – Sept 2024*
Dissertation: Retinal vessel segmentation using U-Net / Dense U-Net on DRIVE dataset — sensitivity at 79.72%, accuracy at 95.71%, Jaccard index at 67.01%, Dice coefficient at 77.84%, and AUC at 97.34%.
Coursework: Deep Learning,CV, NLP, Time Series, Statistical Modelling.

WORK EXPERIENCE

AI/ML Intern · DR.VIKI (Nurtureheal Healthcare Private Limited), Bengaluru May *2026 – Present*

- Developing and optimizing ForceField, an AI-powered movement intelligence platform leveraging Computer Vision, Pose Estimation, and Generative AI–driven coaching for real-time exercise assessment across 12+ exercise modules.
- Building end-to-end AI systems for biomechanical analysis, repetition tracking, performance analytics, PDF reporting, and longitudinal progress monitoring, reducing manual movement evaluation effort by 80%+ through automation.
- Engineering production-ready healthcare AI solutions using MediaPipe, JavaScript, GenAI workflows, Git, GitHub, and Vercel, while implementing personalized insights, voice-guided coaching, and adaptive performance recommendations.

Data Science Intern · AI Variant — AI-Powered Solutions, Bengaluru *Nov 2025 – Feb 2026*

- Delivered two production-style AI MVPs end-to-end: problem framing, data pipelines, ML modelling, REST APIs, dashboards, and live deployment.
- Built an explainable NLP sentiment platform (TF-IDF + SVM, 85.55% accuracy) and a customer segmentation system with scenario simulation; enabled 50% faster analysis cycles and targeted 10–15% potential ROI uplift.
- Established reproducible ML practices: persisted preprocessing artefacts, model calibration, run-based execution, and CI/CD-driven deployments using Python, scikit-learn, FastAPI, Streamlit, and GitHub Actions.

Data Science Intern · Vertexblue Pvt Ltd, India *Jun 2022 – Dec 2022*

- Improved forecasting accuracy by 15% using ARIMA and RNN-based time-series models in Python.

Delivered analytics pipeline contributing to 10%+ cost reduction; reduced manual processing overhead by 30% through automated reporting workflows

Production AI Projects (at Ai Variant):

1. Strategic Intelligence Stack | Lead (Team of 5)

Objective:
Built a production-grade customer segmentation & decision intelligence system to convert raw data into actionable business strategies.

Tools Used:
Python, Pandas, NumPy, Scikit-learn (K-Means), FastAPI, Next.js, Vercel, Render, GitHub Actions (CI/CD)

Business Need:
Static segmentation with low usability, no scenario testing, and high manual effort in analysis.

Solution (Data Science):

- Led team of 5 to develop a deterministic clustering pipeline (K-Means + preprocessing) for consistent segmentation
- Engineered insight layer (revenue mix, discount sensitivity, channel preference) to translate clusters into business actions
- Built a real-time simulation engine for what-if analysis without model retraining
- Implemented run-based architecture for reproducibility and auditability
- Developed executive dashboards for decision-making and reporting
- Set up GitHub-based version control with CI/CD pipelines (GitHub Actions) for automated testing and deployment to Vercel/Render

Impact:

- Improved campaign targeting by ~20–25% and reduced analysis time by ~30–35%, accelerating decision cycles by ~40% (days → hours).
- Increased stakeholder adoption by ~1.5–2x and reduced deployment errors by ~25% via CI/CD automation.

2. ReviewSense AI | Sentiment Intelligence Platform

Objective:

Built a trust-aware NLP system to convert customer reviews into actionable insights and risk signals.

Tools:

Python, Scikit-learn (TF-IDF, classification-SVM, calibration), Pandas, Streamlit, Plotly, Joblib, GitHub Actions (CI/CD)

Business Need:

Unstructured reviews lacked reliable sentiment, risk detection, and prioritization, leading to high manual effort.

Solution:

- Developed TF-IDF + calibrated sentiment model for accurate predictions and achieved 85.55% accuracy & 0.853 F1
- Implemented confidence & risk scoring to flag low-certainty reviews
- Built tricky review detection (negation, mixed sentiment, ambiguity)
- Designed interactive Streamlit dashboard with filters and executive insights
- Set up CI/CD via GitHub Actions for automated testing and deployment

Impact:

- Improved review triaging efficiency by ~25% and reduced manual review effort by ~30% through confidence-based prioritization.
- Increased model trust and adoption by ~1.5x while enhancing risky review detection accuracy by ~20%

DEPLOYED ML SYSTEMS (LIVE DEMOS: PRANAV-GUJJAR-PORTFOLIO.VERCEL.APP)

ForceField — AI Movement Intelligence Platform — *CV · MediaPipe · Pose Estimation · JavaScript · GenAI · Vercel*

- Developed an AI-powered movement intelligence platform providing real-time exercise assessment across 12+ exercises.
- Engineered pose estimation, biomechanical scoring, repetition detection, and adaptive coaching using Computer Vision and MediaPipe.
- Built longitudinal analytics dashboards, PDF reporting, movement history, and voice-guided AI coaching.
- Reduced manual movement assessment effort by 80%+ through automated posture analysis and personalized recommendations.

Retina-AI - Diabetic Retinopathy Screening Platform — *PyTorch · Grad-CAM · FastAPI · Streamlit · GitHub Actions*

- Built a clinical AI screening system with DR / No-DR classification, Grad-CAM explainability, confidence gating, risk stratification, and automated PDF report generation.
- Full workflow: patient registry, role-based authentication, screening, and report management. Deployed with GitHub Actions CI and Ruff linting on Streamlit Cloud.

Identity Vault - Google OAuth Identity Registry — FastAPI · Next.js · PostgreSQL · Google OAuth · Render · Vercel · CI/CD

- Built a deployed identity registry that turns verified Google profiles into searchable internal identity records, reducing manual verification from 2–3 steps to 1 generated-ID lookup.
- Delivered 7 FastAPI endpoints, 5 stored identity fields, PostgreSQL persistence, Vercel/Render/Neon deployment, and GitHub Actions CI with 4 backend tests, Ruff checks, and frontend build validation.

ACADEMIC PROJECTS

Retinal Image Segmentation — MSc Dissertation — *Tensorflow · U-Net · Dense U-Net · DRIVE Dataset*

- Achieved 97.34% ROC-AUC for retinal vessel segmentation; built medical preprocessing pipeline (CLAHE, patch-based learning, augmentation); evaluated with Dice, IoU, Sensitivity, and Specificity.

AI Chatbot — UK Train Ticketing (NLP) — *Python · Random Forest · Gradient Boosting · scikit-learn*

- Intent classification and entity extraction over 400K+ time-series records; benchmarked Linear, RF, GBM, and KNN models; modular reusable preprocessing pipelines with persisted artefacts.